

Diffusion-Limited Reaction Frequencies for One-Dimensionally Migrating Defect Clusters

Abstract

This note derives reaction frequencies for a one-dimensionally (1D) migrating defect cluster interacting with fixed sinks, extends the result to 1D motion with stochastic changes of glide direction (1DR-0), and develops the three geometrical cases for reactions between two purely 1D-mobile clusters. The notation is chosen for cluster-dynamics equations written in atomic fractions. Particular care is taken to distinguish a volumetric bimolecular coefficient, a sink strength, a loss frequency in hertz, and a total reaction-rate density.

1 Definitions and normalization

For BCC iron,

$$\Omega = \frac{a^3}{2}, \quad c_\alpha = \Omega N_\alpha, \quad D_\alpha = a^2 \omega_\alpha. \quad (1)$$

Here N_α is a number density, c_α is an atomic fraction, and ω_α has units s^{-1} . Four related quantities must not be confused:

$$\dot{N}_A = -K_{AB}^{\text{vol}} N_A N_B, \quad [K^{\text{vol}}] = \text{m}^3 \text{s}^{-1}, \quad (2)$$

$$\dot{c}_A = -K_{AB}^{\text{af}} c_A c_B, \quad K_{AB}^{\text{af}} = K_{AB}^{\text{vol}} / \Omega \quad [\text{s}^{-1}], \quad (3)$$

$$\dot{N}_A = -D_A k_A^2 N_A, \quad [k_A^2] = \text{m}^{-2}, \quad (4)$$

$$\dot{c}_A = -\mathcal{K}_A c_A, \quad \mathcal{K}_A = D_A k_A^2 \quad [\text{s}^{-1}]. \quad (5)$$

In this note, *kernel in hertz* means the loss frequency $\mathcal{K}_A$. Such a frequency may depend on the instantaneous sink and reactant populations; it is not necessarily a concentration-independent binary constant.

2 Pure 1D migration to fixed sinks

2.1 Transport analogy: microscopic and macroscopic cross sections

Consider a cluster of class n gliding along a fixed line. A fixed sink of family m captures it when the glide line passes through the sink's projected capture area. In analogy with neutron transport, define

$$\sigma_{n,m} = \pi R_{n,m}^2 \quad (6)$$

as the microscopic capture cross section. The macroscopic cross sections are

$$\Sigma_{n,m} = \sigma_{n,m} N_m, \quad \Sigma_n = \sum_j \Sigma_{n,j} = \sum_j \sigma_{n,j} N_j, \quad (7)$$

and the geometrical mean free path along an uncorrelated line is

$$\lambda_{g,n} = \Sigma_n^{-1}. \quad (8)$$

Thus σ has units m^2 , Σ has units m^{-1} , and λ_g has units m . The symbol κ in the accompanying reaction table is precisely Σ_n .

Unlike a ballistic neutron, a gliding cluster performs a 1D random walk and repeatedly retraces its path. Its absorption time therefore scales as a squared distance divided by D_n , not as a distance divided by a speed.

2.2 First-passage derivation of the sink strength

Let $p(z, t)$ be the probability density on a sink-free chord bounded by absorbing interceptions at $z = 0$ and $z = L$:

$$\frac{\partial p}{\partial t} = D_n \frac{\partial^2 p}{\partial z^2}, \quad p(0, t) = p(L, t) = 0. \quad (9)$$

The mean first-passage time $T(z)$ follows from the backward equation

$$D_n T''(z) = -1, \quad T(0) = T(L) = 0, \quad (10)$$

and hence

$$T(z) = \frac{z(L-z)}{2D_n}. \quad (11)$$

For a uniformly sampled starting point on a chord,

$$\bar{T}(L) = \frac{1}{L} \int_0^L T(z) dz = \frac{L^2}{12D_n}. \quad (12)$$

These equations establish the fundamental scaling $\mathcal{K} \sim D_n/L^2 \sim D_n \Sigma_n^2$.

To resolve the numerical coefficient and the absorbing family, average over the marked Poisson sequence of line interceptions. An interception is labelled m with probability

$$P(m) = \frac{\Sigma_{n,m}}{\Sigma_n}. \quad (13)$$

The self-consistent chemical-rate average over the two bounding interceptions gives the 1D screening length

$$\lambda_{s,n}^{-2} = 6\Sigma_n^2. \quad (14)$$

The coefficient differs from the inverse of $\bar{T}(\lambda_g)$ because the physical ensemble contains a distribution of chord lengths and is weighted by surviving walkers, rather than identical chords of length λ_g . Resolving the terminal interception by its mark gives

$$\begin{aligned} k_{(1),n,m}^2 &= P(m) \lambda_{s,n}^{-2} = 6\Sigma_{n,m} \Sigma_n \\ &= 6\sigma_{n,m} N_m \sum_j \sigma_{n,j} N_j. \end{aligned} \quad (15)$$

This is the Borodin–Barashev 1D sink strength [1, 2]. It follows from line interception, 1D first passage, and the marked-sink ensemble; it is not a Smoluchowski $4\pi RD$ result.

The loss frequency of mobile cluster n to family m is

$$\boxed{\mathcal{K}_{n \rightarrow m}^{(1)} = D_n k_{(1),n,m}^2 = 6D_n \sigma_{n,m} N_m \Sigma_n.} \quad (16)$$

In atomic-fraction variables,

$$\boxed{\mathcal{K}_{n \rightarrow m}^{(1)} = \frac{6D_n \sigma_{n,m}}{\Omega^2} c_m \sum_j \sigma_{n,j} c_j = 24\omega_n \frac{\sigma_{n,m}}{a^4} c_m \sum_j \sigma_{n,j} c_j.} \quad (17)$$

The population equation is

$$\dot{c}_n|_m = -\mathcal{K}_{n \rightarrow m}^{(1)} c_n. \quad (18)$$

2.3 Origin and consequences of the R^4 law

For one monodisperse family, with $\sigma = \pi R^2$,

$$k_{(1)}^2 = 6\sigma^2 N^2 = 6\pi^2 R^4 N^2, \quad \mathcal{K}^{(1)} = 6\pi^2 D_n R^4 N^2. \quad (19)$$

The two factors of R^2 have different roles. One cross section selects the family responsible for terminal capture; the second fixes the spacing between all line interceptions and hence the distance explored by 1D diffusion. This produces a loss frequency quadratic in sink density and proportional to R^4 .

For a spherical cavity containing m vacancies,

$$R_m = a \left(\frac{3m}{8\pi} \right)^{1/3}, \quad (20)$$

so

$$k_{(1),m}^2 \propto m^{4/3} N_m^2. \quad (21)$$

The result assumes dilute, statistically independent sinks and a fixed glide direction. Correlated positions, overlap of projected cross sections, finite reaction probabilities, and orientation-dependent nonspherical sinks require modification.

3 1D migration with rotations: the 1DR–0 kernel

Suppose cluster n makes 1D flights and changes glide direction after a characteristic flight length ℓ_n . Its trajectory is 1D below ℓ_n but becomes isotropic at sufficiently large scales. Let $k_{(1),n,m}^2$ be given by [equation \(15\)](#), and let $k_{(3),n,m}^2$ be the conventional 3D sink strength for the same sinks. The Trinkaus–Heinisch master curve is

$$k_{(1R),n,m}^2 = y_{n,m} k_{(1),n,m}^2, \quad (22)$$

$$y_{n,m} = \frac{1}{2} \left(1 + \sqrt{1 + 4x_{n,m}^2} \right), \quad (23)$$

$$x_{n,m} = \left[\frac{\ell_n^2 k_{(1),n,m}^2}{12} + \frac{k_{(1),n,m}^4}{k_{(3),n,m}^4} \right]^{-1/2}. \quad (24)$$

This x is the reciprocal of another common convention; equations (23) and (24) must be copied together [3, 2]. The hertz-valued kernel is

$$\boxed{\mathcal{K}_{n \rightarrow m}^{(1R)} = D_n y_{n,m} k_{(1),n,m}^2.} \quad (25)$$

3.1 Pure-1D limit

If $\ell_n \sqrt{k_{(1)}^2} \gg 1$, absorption normally precedes rotation. Then $x \rightarrow 0$, $y = 1 + x^2 + O(x^4)$, and

$$\boxed{\mathcal{K}^{(1R)} \longrightarrow D_n k_{(1)}^2}. \quad (26)$$

3.2 Frequent-rotation, 3D limit

If the flight length is short compared with the absorption length, the first term in [equation \(24\)](#) vanishes and

$$x \longrightarrow \frac{k_{(3)}^2}{k_{(1)}^2}. \quad (27)$$

In the usual dilute-sink regime $k_{(3)}^2 \gg k_{(1)}^2$, hence $x \gg 1$, $y = x + 1/2 + O(x^{-1})$, and

$$\boxed{\mathcal{K}^{(1R)} \longrightarrow D_n k_{(3)}^2}. \quad (28)$$

Thus one frequency recovers both limiting sink strengths. The definition of ℓ_n (mean, projected, or RMS flight length) and the convention used for the long-time diffusivity must match the cited master curve.

4 Reactions between two purely 1D-mobile clusters

Let clusters A and B have glide axes $\mathbf{e}_A, \mathbf{e}_B$, diffusivities D_A, D_B , and mutual capture distance $R = R_A + R_B$. There are three geometrically distinct cases.

4.1 Variant I: collinear or coincident glide lines

When both species sample the same line family, the other mobile population also sets the interception spacing. Applying the marked-interception result symmetrically gives the collinear rate density proposed by Adjanor [\[4\]](#):

$$\mathcal{R}_{\parallel} = 6\pi^2 R^4 (N_A^2 N_B D_B + N_B^2 N_A D_A). \quad (29)$$

Dividing by N_A gives the loss frequency of an A cluster:

$$\boxed{\mathcal{K}_{A|B}^{\parallel} = 6\pi^2 R^4 (D_B N_A N_B + D_A N_B^2)}. \quad (30)$$

Equivalently,

$$\boxed{\mathcal{K}_{A|B}^{\parallel} = \frac{6\pi^2 R^4}{\Omega^2} (D_B c_A c_B + D_A c_B^2)}. \quad (31)$$

The kinetics is cubic in populations, so this is an effective many-particle frequency, not a concentration-independent binary constant. Strictly parallel but distinct lines do not react under perfect 1D motion unless their separation is smaller than R ; lattice-plane occupancy must then be included.

4.2 Variant II: nonparallel intersecting glide lines

Let s_A, s_B be displacements along two nonparallel lines that intersect. Near the intersection,

$$r^2 = s_A^2 + s_B^2 - 2s_A s_B \cos \theta, \quad (32)$$

and the pair probability obeys

$$\frac{\partial p}{\partial t} = D_A \frac{\partial^2 p}{\partial s_A^2} + D_B \frac{\partial^2 p}{\partial s_B^2}, \quad p = 0 \quad (r = R). \quad (33)$$

There are two independent diffusive coordinates, so the encounter maps to a two-dimensional first-passage problem. For equal diffusivities, the standard midpoint mapping gives the long-time result

$$\mathcal{R}_\times \simeq K_\times^{\text{vol}}(t) N_A N_B, \quad (34)$$

$$K_\times^{\text{vol}}(t) = 2\pi D R \alpha(t), \quad (35)$$

$$\alpha(t) \simeq \frac{4}{\ln(4Dt/\pi R^2)}. \quad (36)$$

The slowly varying logarithm is characteristic of 2D diffusion [5, 4]. The hertz-valued loss kernel is

$$\boxed{\mathcal{K}_{A|B}^\times(t) = 2\pi D R \alpha(t) N_B = \frac{2\pi D R}{\Omega} \alpha(t) c_B.} \quad (37)$$

This contribution is approximately quadratic in concentration. For $D_A \neq D_B$ or nonorthogonal axes, [equation \(33\)](#) transforms to diffusion toward an ellipse. The diffusion-tensor transformation or a validated anisotropic correlation must be used; replacing D by $D_A + D_B$ is not generally valid [6, 4, 7].

4.3 Variant III: nonparallel skew, non-coplanar glide lines

Let h be the minimum separation of the two infinite glide lines. At their closest points, contact requires $h^2 + q^2 \leq R^2$, where q is distance in the two-dimensional pair-coordinate plane. Therefore

$$R_{\text{eff}}(h) = \begin{cases} \sqrt{R^2 - h^2}, & h < R, \\ 0, & h \geq R. \end{cases} \quad (38)$$

The intersecting-line derivation applies with $R \rightarrow R_{\text{eff}}$:

$$\boxed{\mathcal{K}_{A|B}^{\text{skew}}(t, h) = 2\pi D R_{\text{eff}}(h) \alpha[t; R_{\text{eff}}(h)] N_B.} \quad (39)$$

For a lattice population, average over admissible plane separations and glide variants. Since α also depends on radius, the consistent average is

$$\overline{R_{\text{eff}} \alpha} = \sum_h P(h) R_{\text{eff}}(h) \alpha[t; R_{\text{eff}}(h)], \quad (40)$$

not necessarily $\bar{R}_{\text{eff}} \alpha(t; \bar{R}_{\text{eff}})$.

4.4 Orientation-averaged 1D–1D kernel

If v unoriented glide-axis variants are equally populated, the fraction of non-collinear pairs is

$$f_v = \frac{v-1}{v}. \quad (41)$$

For the four unoriented $\langle 111 \rangle$ axes in BCC iron, $f_v = 3/4$. Separating intersecting and skew nonparallel pairs by conditional probability $p_\times$ gives

$$\mathcal{K}_{A|B}^{1D-1D} = (1 - f_v)\mathcal{K}_{A|B}^{\parallel} + f_v \left[p_\times \mathcal{K}_{A|B}^{\times} + (1 - p_\times) \langle \mathcal{K}_{A|B}^{\text{skew}} \rangle \right]. \quad (42)$$

This displays what is hidden when only the $6\pi^2 R^4$ term is quoted. In a dilute system the approximately quadratic non-collinear contribution commonly exceeds the cubic collinear contribution. Spatial correlations, rotations, finite cell size, and unequal occupation of glide variants change the weights.

5 Implementation summary

For each mobile class n and fixed family m :

1. Compute $\sigma_{n,m}$ and $\Sigma_n = \sum_j \sigma_{n,j} N_j$.
2. Form $k_{(1),n,m}^2 = 6\sigma_{n,m} N_m \Sigma_n$.
3. If rotations occur, compute x, y , and $\mathcal{K}_{n \rightarrow m}^{(1R)} = D_n y k_{(1),n,m}^2$.
4. For two purely 1D-mobile classes, identify the collinear, intersecting, or skew geometry and use the corresponding frequency.
5. Apply orientation and lattice-plane weights before summing frequencies.
6. For an $A + A$ reaction, use the conventional factor $1/2$ in the number of events; two A objects are removed by each event.

Scope. The formulas assume diffusion-controlled, perfectly absorbing encounters in a homogeneous dilute medium. They exclude reaction barriers, elastic drift, correlated cascade positions, emission, and rotation changes caused by trapping or stress. Such effects should modify the capture probability, geometry, or migration model explicitly rather than being hidden in R .

References

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